

CSE110_Sec21_Fall25_Quizzes

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Question - 1: [20 points]

A shipping company needs to deliver products to a destination. The truck that will carry the products has a capacity of **X kilograms**. If the product weight exceeds this capacity, it cannot be shipped. If the weight is within capacity, the shipping charge should be calculated. The shipping charge is the product of the **weight charge** and the **distance**.

- For product weight **up to 20 kg**, the weight charge is **5 tk per kg**.
- For each kilogram **beyond 20 kg**, there is a **3 tk extra charge** with every additional kg of weight along with the **existing 5 tk charge**.

Total shipping charge = weight charge × distance.
Take the truck capacity (in kg), product weight (in kg), and distance (in km) as user inputs and write a Java program to solve this problem.

Sample Input	Sample Output
Please enter the truck capacity (kg): 100.0 Please enter the product weight (kg): 10.0 Please enter the distance (km): 50.0	Total shipping charge: 2500.0 tk
Please enter the truck capacity (kg): 200.0 Please enter the product weight (kg): 25.0 Please enter the distance (km): 30.0	Total shipping charge: 4200.0 tk
Please enter the truck capacity (kg): 50.0 Please enter the product weight (kg): 60.0 Please enter the distance (km): 80.0	Product cannot be shipped. Exceeds truck capacity.

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSE110: PROGRAMMING LANGUAGE I

QUIZ II

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Question - 1: [14 points]

A university cafe called Mojaar Khabar wants to give students a discount for 7 consecutive days based on the profit they are making the previous day.

Calculation for the discount :

- (Summation of the odd digits of previous day profit % 10). If the odd digit is divisible by 3, it should be squared before summation.

Write a Java program which will take the profit as user input for 7 days and find the discount percentages. **Note:** You cannot use Arrays.

Sample Input	Sample Output
Enter profit for Day 1: 619635 Enter profit for Day 2: 123456 Enter profit for Day 3: 7398135 Enter profit for Day 4: 1905678 Enter profit for Day 5: 234561 Enter profit for Day 6: 8532 Enter profit for Day 7: 10935	No discount for the first day, sorry! Discount for Day 2 : 6 % Discount for Day 3 : 5 % Discount for Day 4 : 2 % Discount for Day 5 : 4 % Discount for Day 6 : 5 % Discount for Day 7 : 4 %

Explanation:

As there is no previous day before day1, there is no discount for day 1. For the rest of the days, there will be a discount. Ex: For day 2, discount is based on the profit obtained on day 1. The odd digits in day 1 profit are 1,9,3,5. **9 and 3** will be squared before adding as they are divisible by 3 and the rest of the odd digits will be added as it is and modded by 10. $(1 + 9^2 + 3^2 + 5) \% 10 = 96 \% 10 = 6 \rightarrow$ discount percentage for day 2.

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Question - 2: [6 points]

Write the outputs (5 in number) in the output box by tracing the following code. You must show a proper trace table and calculation, for which 1 mark will be allocated.

1	public class Tracing_Quiz2 {	Output:
2	public static void main(String[] args) {	
3	int a = 4, b = 9, c = 1;	
4	double d = 2.5;	
5	while (true) {	
6	++a;	
7	double val;	
8	if (a == 7) {	
9	continue;	
10	}	
11	if (a % 3 == 0) {	
12	b = (b / a) + c++;	
13	System.out.println(b); }	
14	else {	
15	val = (b / a) + c * ((int)d);	
16	System.out.println(val); }	
17	c -= - a + 1;	
18	if (a == 10) {	
19	break;	
20	}	
21	}	
22	}	
23	}	

1.

2.

3.

4.

5.

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CSE110: PROGRAMMING LANGUAGE I

QUIZ III

NAME:	ID:	SECTION:
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Question - 1: [20 points]

You are given an array of integers from which you have to check if the consecutive differences between adjacent numbers are even or odd. If the difference is even, you need to store true in a **boolean array**; else, store false. Print the elements of the boolean array.

Note: You cannot use any in-built method to solve this.

Given array	Given Output
int [] arr = {5, 6, 4, 1, 3, 8}	[false, true, false, true, false]
int [] arr = {6, 1, 7, 2, 4, 6, 7}	[false, true, false, true, true, false]

Explanation:

Difference between 6 and 1: 5 -> Odd

Difference between 1 and 7: 6 -> Even

Difference between 7 and 2: 5 -> Odd

Difference between 2 and 4: 2 -> Even

Difference between 4 and 6: 2 -> Even

Difference between 6 and 7: 1 -> Odd

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Tracing

Set 1

3.0

6

22.0

18

55.0

Set 2:

9.0

17.0

11

46.0

23

Set 3:

6

9.0

16

26.0

35.0

Set 4:

15.0

10

18.0

24.0

16

Code:

import java.util.Scanner;

public class MojaarKhabar{

public static void main(String[] args) {

Scanner sc = new Scanner(System.in);

int discount =0;

for (int day = 1; day<=7; day++)

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Set A:

1. 5

2. 2

3. 8

4. 7

5. 7

6. 8

7. 10

8. 1

9. True

Set B:

1. 6

2. 3

3. 4

4. 2

5. 1

6. 1

7. 2

8. -5

9. True

Set C:

1. 5

2. 2

3. 12

4. 18

5. 25

6. 33

7. 42

8. 39

9. false

Set D:

1. 3

2. 4

3. 9

4. 16

5. 24

6. 33

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5

6



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